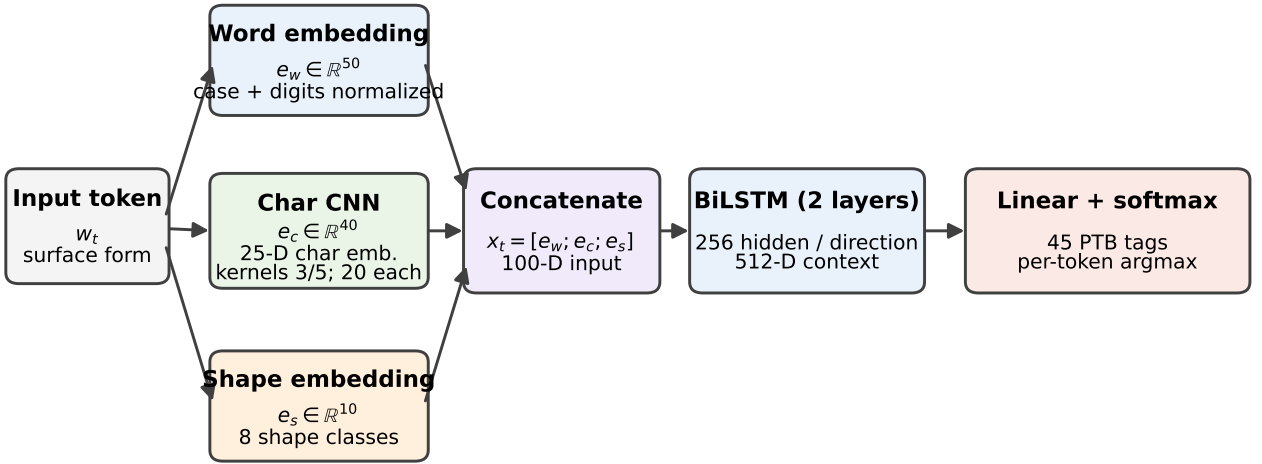
Selected model: 50-D word + 40-D character + 10-D shape
Word embedding
$e_w \in \mathbb{R}^{50}$
case + digits normalized
Input token
w_t
surface form
Char CNN
$e_c \in \mathbb{R}^{40}$
25-D char emb.
kernels 3/5; 20 each
Shape embedding
$e_s \in \mathbb{R}^{10}$
8 shape classes
Concatenate
$x_t = [e_w; e_c; e_s]$
100-D input
BiLSTM (2 layers)
256 hidden / direction
512-D context
Linear + softmax
45 PTB tags
per-token argmax
No pretrained vectors • final checkpoint: 4,067,274 parameters • label smoothing only during training